Stuyvesant Alumni Perspectives:
Milan Haiman, Class of 2019

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In his senior year as a math major at MIT, Milan Haiman (Class of 2019) shares his memories of Stuy, his love for combinatorics, and his varied math experiences, including winning a Gold medal (8th place) for Hungary in the 2019 International Mathematics Olympiad (IMO) in the United Kingdom.

1 The biggest factor in Milan's success in competing in and teaching math? Going to Stuy

Although Milan did not participate in math competitions before high school, he attended New York Math Circle from fifth grade through middle school. One of his NYMC teachers was Mr. Kats, who would eventually also teach one of Milan's calculus and two of his Math Team classes and become one of his favorite teachers. During middle school, Milan also attended MathPath and in high school, he participated in MathILy, Canada/USA Mathcamp, Math-M-Addicts, and MOP (Mathematics Olympiad Program), but he was most impacted by his four years at Stuy. He particularly enjoyed training and competing with the Math Team, where he served as a captain.

"I made lots of friends on the Math Team," he emphasized. "If I hadn't done that, then I don't think I would have liked math as much and done as much math in high school and probably would not have been as interested in college, etc." Although Milan's specific interests in math have changed since high school, he remarked, “You can keep changing how you think about [math], but as long as there’s people whom you like working with, I think that’s probably the most important thing.” Milan especially enjoyed all the New York City Math Team trips to competitions such as ARML (American Regional Mathematics League) and NYSML (New York State Mathematics League), and as an alumnus, he reunited with the teams and coaches when they traveled to Cambridge, Massachusetts to compete in HMMT (Harvard-MIT Mathematics Tournament).

As a senior Math Team captain at Stuy, Milan also gained valuable teaching experience, leading Math Team classes every morning and organizing practices for an entire year. “I got to choose whatever I wanted to teach,” he recalled fondly. Later, he would apply this experience in his college and summer employment. “Overall, I think I really like teaching,” he shared. “I definitely want to continue teaching in grad school.”

2 Before, during, and after the IMO

When asked for a fun fact about himself, Milan replied that he is half Indian and half Hungarian. But the whole Milan represented Hungary in the 2019 International Mathematical Olympiad, which was held in the United Kingdom in the city of Bath. He described some of his training before the global competition, saying, “The summer before my senior year, I went to some camps in Hungary.” Then in the winter, he attended a joint Hungary-UK math camp for 20 British students and 20 Hungarian students, which included interesting lectures and several practice Olympiad exams.

Milan then traveled for 10 hours from New York to Bucharest to represent Hungary at RMM (Romanian Master of Mathematics). He thought it was interesting that the travel time for the three other students who represented Hungary was longer than his. “[Bucharest] is in the South of Romania and Budapest is in the north of Hungary, so it’s around 14 hours on the train,” he explained. His adventure included sleeping in the same room as his three teammates on very squeaky beds and searching for vegetarian food options for himself. As for the contest itself, Milan recounted an interesting turn of events on Day 1 of the two-day competition. “Because of an unexpected incident, the Day 2 problems had to be changed,” Milan described. “So the problems on the second day were not as good. I later heard about what was supposed to be on that day, and it seemed much more interesting.”

After one additional training camp in Hungary, Milan headed for Bath, which is about a two-hour drive from London, to compete in the IMO. He cherished meeting students from all over the world and appreciated that it was well-organized, including the availability of “really nice fruits.” He also had the opportunity to visit famous historical monuments, recollecting, “After the contest days of the IMO, the students get to go on excursions while the papers are being graded. So, I got to see Stonehenge, which was pretty cool! It was quite big and I have no idea how people made it so long ago.” During another trip, Milan and his fellow competitors enjoyed “see[ing] the actual baths” in Bath.

Milan ultimately earned a Gold medal, scoring 40 out of 42 points, ranking an impressive 8th place in the world. When asked at what point he realized that he had the potential to perform so well on an international level, he replied, “I think probably after the IMO was over.” He confided, “I also think I was quite lucky with the problems on the IMO. My best subject was combinatorics and then after that was geometry, but mostly because I knew how to bash the geometry problems, not solve them how you are supposed to do. The way the IMO works is there's three problems on each day, so six total, and then each day [the problems] increase in difficulty. And there's a mix of subjects: algebra, number theory, combinatorics, and geometry. But there's obviously going to be some subjects for the harder problems and some for the easier problems. So the first problems [on each day], #1 and #4, which are the two easiest, were algebra and number theory and #2 and #3 were geometry and combinatorics, and then #5 and #6 were combinatorics and geometry. So I still think I was pretty lucky, and if the tests were different, then I probably wouldn't have done as well."

Milan continued his involvement with Olympiad math after high school. The summer after junior year of college, he worked as a counselor at MOP, teaching mostly on subjects in combinatorics but also on topics in geometry, like inversion. But he reflected, "Maybe the thing I'm happiest about is having a problem on the IMO." Below is Problem 3 of the 2020 IMO which Milan wrote (except he originally framed the problem using coins rather than pebbles):

> *There are $4n$ pebbles of weights $1, 2, 3, \ldots, 4n$. Each pebble is colored in one of n colors and there are four pebbles of each color. Show that we can arrange the pebbles into two piles so that the following two conditions are both satisfied:*
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> – The total weights of both piles are the same.
> – Each pile contains two pebbles of each color.

"There are all the students who go to the IMO, and the IMO problems are preserved forever, so lots of people will look at them," he explained. "So having a nice problem that I liked and having that shared with everyone, I think that's pretty cool."

3 Living, learning, researching, and teaching at MIT

When asked about what makes an MIT education unique, Milan asserts, "The main thing that makes MIT different from other colleges is how the dorms work. At some colleges, I think the culture is that you don't really spend much time in your dorm, and you hang out in other places, but at MIT, each dorm has different [groups of] people, so you have potentially similar kinds of people in the same dorm. You more or less pick which one you want to be in, so there's some self-selection. The dorm I'm in is called Next House; I've made a lot of friends here. Also, the dorms aren't divided [among] class years. When I was a freshman, the person across the hall from me was a junior, and the person next to me was a senior, so you're just mixed in with all the different years, and you can ask for lots of advice from them. I think most colleges will have a freshman dorm. There's probably advantages to both [systems], but I think definitely one of the reasons I wanted to go to MIT was how the dorms work."

In addition to completing his undergraduate coursework, Milan is specifically interested in researching "extremal combinatorics." He has already begun pursuing this passion, most notably through Summer Program in Undergraduate Research (SPUR) at MIT. His research, with his mentor Yannick Yao, entitled "Irreducibility of Generalized Permutahedra, Supermodular Functions, and Balanced Multisets" won the Hartley Rogers Jr. Prize in 2022. According to the Roger's Prize website, Milan's paper "makes a serious contribution to the study of irreducible generalized permutohedra, establishing a double exponential upper bound for the number of such objects in dimension n." From the Duluth Research Experiences for Undergraduates (REU) 2021 summer program, Milan's research was entitled "The Dimension of Divisibility Orders and Multiset Posets."

In his earliest undergraduate research from 2020 to 2021, Milan was supervised by Professor Yufei Zhao while researching "Graphs with high second eigenvalue multiplicity."

Milan also found several opportunities to teach math during his college years. He volunteered to lecture on Catalan numbers at MIT Splash during his freshman and senior years of college, up to 80 students at one time. Milan also worked as a counselor the summer after freshman year college at Canada/USA Mathcamp and as an instructor (with another MIT student) for the Ghana IMO camp in January of his sophomore year of college, both of which were limited to online participation during the pandemic. Being employed as a teaching assistant for a large class called "Probability and Random Variables" during senior year of college allowed Milan to teach weekly recitations and hold office hours. "As a grad student, I'm sure there will be lots of opportunities to do [more] teaching," he predicted.

4 A few tips for current Stuy students

- Use your time wisely.

  "I know a lot of Stuy students will lose sleep for small improvements on their grades. I definitely don't think that's worth it. Getting sleep is a good use of time!"

- Think about what you're doing.

  "Don't just do it because that's what you were doing before and that's what everyone else is doing. You should enjoy what you're doing."

- When writing a math problem, make it something you want to think about just for the sake of thinking about it.

  "So I probably have stronger opinions than the average person about what makes a good problem. I think it should have a nice statement, and it should have a nice solution. The solution should really be about coming up with an idea. And then once you've come up with that idea, you should be able to solve the problem easily. It shouldn't be about looking at the problem, having a general sense of what's going to happen, but then having to do a bunch of calculations to figure out the solution. When I read the problem, I want it to be something I would actually want to think about."

- If you're in Mr. Chew's senior Math Team class, understand isosceles tetrahedrons.

  "He'll want you to learn a trick to deal with 3D geometry."